ii) Electron withdrawing substituents increases the acidity

The electron – withdrawing substituents decrease the negative charge on the carboxylate ion and stabilize it. In such cases, the loss of proton becomes relatively easy.

Acidity increases with increasing electronegativity of the substituents. For example, the acidity of various halo acetic acids follows the order

$$F-CH_2-COOH > Cl-CH_2-COOH > Br-CH_2-COOH > I-CH_2-COOH$$

Acidity increases with increasing number of electron – withdrawing substituents on the α - carbon. For example

$$Cl_3C-COOH > Cl_2CH-COOH > ClCH_2COOH > CH_3COOH$$

The effect of various, electron withdrawing groups on the acidity of a carboxylic acid follows the order,

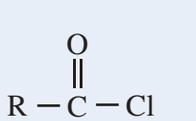
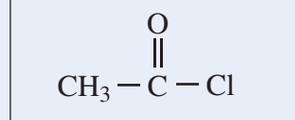
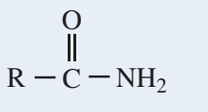
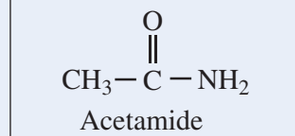
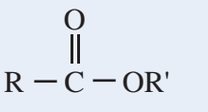
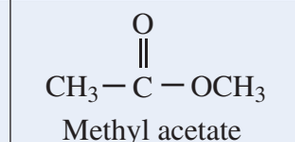
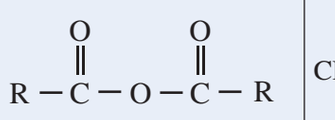
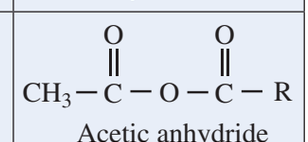
$$-NO_2 > -CN > -F > -Cl > -Br > -I > -Ph$$

The relative acidities of various organic compounds are

$$RCOOH > ArOH > H_2O > ROH > RC \equiv CH$$

12.14 Functional derivatives of carboxylic acids

Compounds such as acid chlorides, amides, esters etc., are called carboxylic acid derivatives because they differ from a carboxylic acid only in the nature of the group or atom that has replaced the -OH group of carboxylic acid.

Group replacing - OH	Name	Structure	Example
–Cl	Acid chloride		 Acetyl chloride
$-NH_2$	Acid amide		 Acetamide
–OR'	ester		 Methyl acetate
–OOCR	Acid anhydride		 Acetic anhydride

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